[Move to Section §3.4]

§(5.3) Generation/Recombination Currents

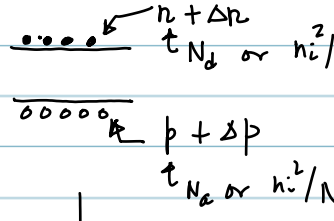
Excess carriers: removed by various mechanisms

$$R - G = \frac{np - n_i^2}{\tau_n(p + p_t) + \tau_p(n + n_t)} \quad \text{"SRH"}$$

$$= X \cdot (np - n_i^2)$$

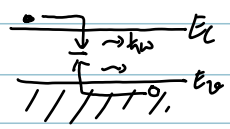
- B (radiative)
- $C_n n + C_p p$ (Auger, non-rad)
- $\frac{1}{\tau_p(n+n_t) + \tau_n(p+p_t)}$ (SRH, non-rad)

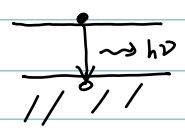
Majority Carriers:
(Dielectric relaxation!)
(flow out to contacts!) $\tau_d \approx \frac{\varepsilon_s}{\sigma}$
$[1/\tau_d \approx \frac{\sigma}{\varepsilon_s}]$

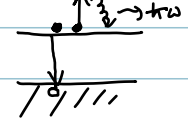


Minority Carriers ⇐ (Recombination)

- (DEFECT): $\frac{1}{\tau_{SRH}} = An$ (non-radiative)
- (INTRINSIC): $\frac{1}{\tau_{rad}} = Bn^2 = B(n \cdot p - n_i^2)$ (Radiative)
- (INTRINSIC): $1/\tau_{Aug} = Cn^3 = C_n n^2 p + C_p n p^2 \propto (C_n n + C_p p)(np - n_i^2)$ (Non-radiative) — Auger–Meitner process.



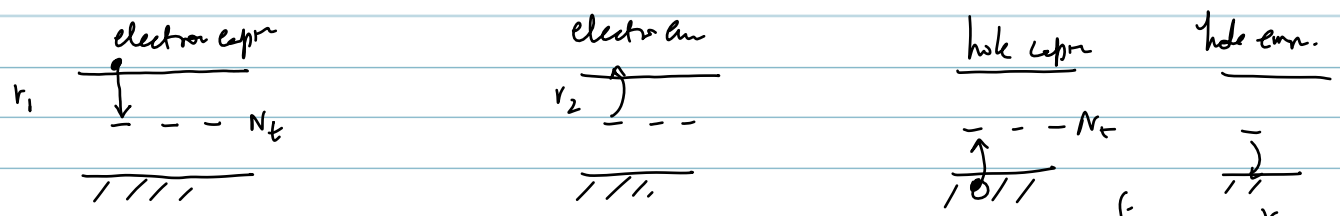




$\frac{1}{\tau_{SRH}} \neq$

$$\left\{ \text{for LED's: } \eta^{IQE} = \frac{\text{Radiative}}{\text{All}} = \frac{Bn^2}{An + Bn^2 + Cn^3} < 1 = \frac{1/\tau_{rad}}{1/\tau_{rad} + 1/\tau_{nr}} \right\}$$

SRH Processes



$r_1 = \text{rate} = n \cdot N_t \cdot (1 - f(E_t))\, v_{th} \cdot \sigma_n$ $\qquad r_2 = N_t f(E_t) \cdot e_n$ $\qquad r_3 = p N_t f \, v_{th} \sigma_p$ $\qquad r_4 = N_t(1-f) e_p$

@ Thermal Equ $\quad r_1 = r_2$

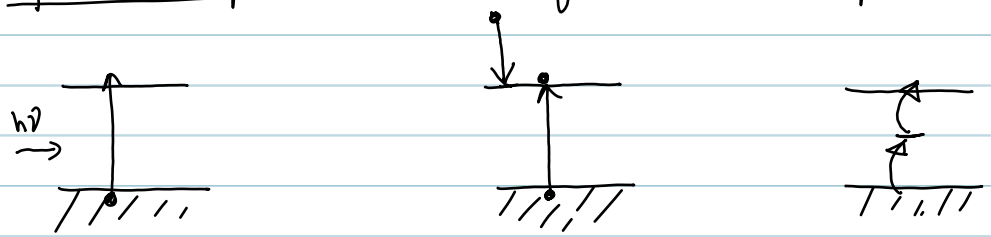
$$e_n = n\, v_{th} \cdot \sigma_n \cdot \frac{1 - f(E_t)}{f(E_t)} = n \cdot v_{th} \cdot \sigma_n \cdot \frac{1 - f(E_t)}{f(E_t)}$$

$$= v_{th} \sigma_n \cdot n_i \exp\left(\frac{E_t - E_i}{kT}\right) \rightsquigarrow \text{Derivation later.}$$

$e_p = v_{th} \sigma_p n_i (e^{\ldots} - 1)$

$$SRH\ (R - G) = r_1 - r_2 = r_3 - r_4$$

Generation processes: Reverse of recombination processes, when ($np < n_i^2$)

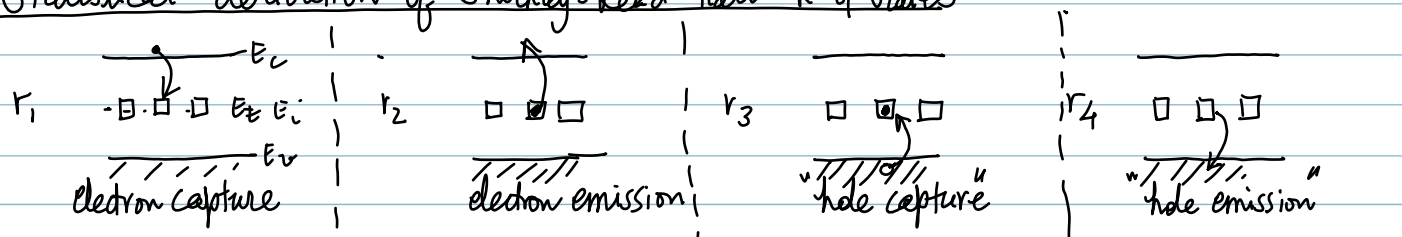


band-to-band radiative — Solar cells

Impact ionization — Breakdown

Trap-assisted generation — ~~Low-biased "avalanche"~~ Reverse bias leakage in pn junction diodes.

Statistical derivation of Shockley-Read-Hall R-G rates



$r_1 = \sigma_n N_t v_{th} n (1-f)$ $\qquad$ $r_2 = e_n N_t f$ $\qquad$ $r_3 = \sigma_p N_t v_{th} p f$ $\qquad$ $r_4 = e_p N_t (1-f)$

Equal @ thermal Eq^m; $\left[f_i = \frac{1}{1+\exp\left(\frac{E_t - E_i}{kT}\right)} \quad n = n_i \quad E_F = E_i \right]$

Equal @ thermal Eq^m. $\left[f = f_i = \frac{1}{1+e^{\frac{E_t - E_i}{kT}}} \quad p = n_i \right]$

$\Downarrow$

$$\sigma_n N_t v_{th} n_i (1-f_i) = e_n N_t f_i$$

$$\Rightarrow \quad e_n = \sigma_n v_{th} n_i \left(\frac{1}{f_i} - 1\right)$$

$$e_n = \sigma_n v_{th} \underbrace{n_i \exp\left(\frac{E_t - E_i}{kT}\right)}_{n_t}$$

$$\boxed{e_n = \sigma_n v_{th} n_t}$$

$$n_t = n_i e^{\frac{E_t - E_i}{kT}}$$

$$p_t = n_i e^{-\left(\frac{E_t - E_i}{kT}\right)}$$

$$n_t p_t = n_i^2$$

$$\sigma_p N_t v_{th} n_i f_i = e_p N_t (1-f_i)$$

$$\Rightarrow \quad e_p = \sigma_p v_{th} n_i \cdot \left(\frac{1}{\frac{1}{f_i} - 1}\right)$$

$$= \sigma_p v_{th} \underbrace{n_i \exp\left(-\frac{E_t - E_i}{kT}\right)}_{p_t}$$

$$\boxed{e_p = \sigma_p v_{th} p_t}$$

Net recombination rate is $R - G = -\frac{\partial(\Delta n)}{\partial t} = -\frac{\partial(\Delta p)}{\partial t} = r_1 - r_2 = r_3 - r_4$.

@ Non-equilibrium, ($f \neq f_i$),

$$r_1 - r_2 = \sigma_n N_t v_{th} n (1-f) - \sigma_n N_t v_{th} n_t \cdot f = r_3 - r_4 = \sigma_p N_t v_{th} p \cdot f - \sigma_p N_t v_{th} p_t (1-f)$$

$$\sigma_n \cancel{N_t} \cancel{v_{th}} [n(1-f) - n_t f] = \sigma_p \cancel{N_t} \cancel{v_{th}} [pf - p_t(1-f)]$$

$$\sigma_n [n - nf - n_t f] = \sigma_p [pf - p_t + p_t f]$$

$$\sigma_n n + \sigma_p p_t = \sigma_n f (n + n_t) + \sigma_p f (p + p_t)$$

$$\boxed{f = \frac{\sigma_n n + \sigma_p p_t}{\sigma_n (n + n_t) + \sigma_p (p + p_t)}}, \quad \boxed{1 - f = \frac{\sigma_n n_t + \sigma_p p}{\sigma_n (n + n_t) + \sigma_p (p + p_t)}}$$

$$\Rightarrow r_1 - r_2 = N_t v_{th}\sigma_n\left[n(1-f) - n_t f\right] = N_t v_{th}\sigma_n \cdot \left[\frac{(\cancel{\sigma_n n \cdot n_t} + \sigma_p p \cdot n) - (\cancel{\sigma_n n \cdot n_t} + \sigma_p \overbrace{p_t \cdot n_t}^{n_i^2})}{\sigma_n(n+n_t) + \sigma_p(p+p_t)}\right]$$

$$\Rightarrow -\frac{\partial}{\partial t}(\Delta n) = -\frac{\partial}{\partial t}(\Delta p) = R - G = \frac{N_t v_{th}\sigma_n\sigma_p\,(np - n_i^2)}{\sigma_n(n+n_t) + \sigma_p(p+p_t)}$$

$$= \frac{np - n_i^2}{\underbrace{\frac{1}{N_t v_{th}\sigma_p}}_{\tau_p}(n+n_t) + \underbrace{\frac{1}{N_t v_{th}\sigma_n}}_{\tau_n}(p+p_t)}$$

$$\Rightarrow \boxed{-\frac{\partial}{\partial t}(\Delta n) = -\frac{\partial}{\partial t}(\Delta p) = R - G = \frac{np - n_i^2}{\tau_p(n+n_t) + \tau_n(p+p_t)}}$$ ← [SRH recombination mechanism]

Apply this to the depletion region of a pn junction diode →

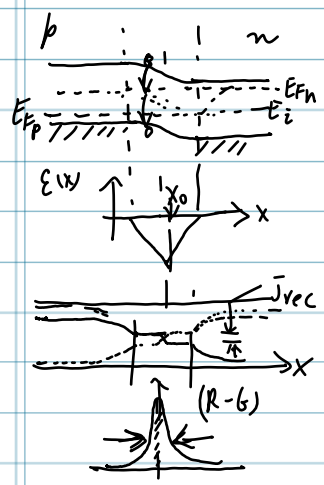
Ⓐ $\boxed{\text{FWD BIAS:}}$

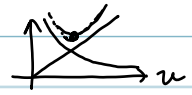
$np \gg n_i^2$ assume $\tau_p = \tau_n = \tau_0 \Rightarrow D^r = \tau_0\left[n + p + 2n_i\cosh\left(\frac{E_t - E_i}{kT}\right)\right]$

$N^r = np = n_i^2 \exp\left(\frac{qV}{kT}\right)$ $\qquad = \tau_0\left[n + \frac{n_i^2}{n}e^{\frac{qV}{kT}} + 2n_i\cosh\left(\frac{E_t - E_i}{kT}\right)\right]$

$$\Rightarrow -\frac{\partial(\Delta n)}{\partial t} = -\frac{\partial}{\partial t}(\Delta p) = R - G = \frac{n_i^2\exp(qV/kT)}{\tau_0\left[\underset{n(x)}{n} + \frac{n_i^2\exp\left(\frac{qV}{kT}\right)}{\underset{n(x)}{n}} + 2n_i\cosh\left(\frac{\underset{E_t(x)}{E_t} - \underset{E_i(x)}{E_i}}{kT}\right)\right]}$$

if $E_t = E_i \Rightarrow \cosh(0) \approx 1$.





$$\frac{\partial}{\partial u}\left(u + \frac{A}{u}\right) = 1 - \frac{A}{u_0^2} = 0 \Rightarrow u_0 = \sqrt{A}$$ is the minimum point.

$$\Rightarrow (R-G)_{max} \cong (R-G)_{u_0} \approx \frac{n_i^2\exp(qV/kT)}{\tau_0\left[n_i\exp\left(\frac{qV}{2kT}\right)\right]} \approx \frac{n_i}{\tau}\exp\left(\frac{qV}{2kT}\right)$$

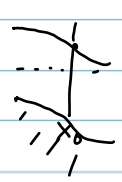
$$J_{rec} \simeq q\int(R-G)\,dx \approx \frac{qn_i}{\tau_0}\exp\left(\frac{qV}{2kT}\right)(\text{Length})$$??

$$n(x) \simeq n_{max}\exp\left(\frac{q\mathcal{E}(x_0)x}{kT}\right)$$

$$p(x) \simeq n_{max}\exp\left(-\frac{q\mathcal{E}(x_0)x}{kT}\right)$$

Characteristic Length $\approx \pi\frac{(kT/q)}{\mathcal{E}(x_0)}$

$\mathcal{E}$ (Electric field $\mathcal{E}$ max recomb plane)



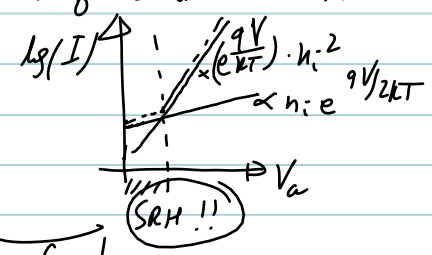
$$\Rightarrow -\frac{\partial(\Delta n)}{\partial t} \approx \frac{n_i^2\exp(qV/kT)}{2\tau_0 n_i\exp\left(\frac{qV}{2kT}\right)\left[\cosh\left(\frac{q\mathcal{E}(x_0)x}{kT}\right)\right]}$$

$$-\frac{\partial(\Delta n)}{\partial t} = -\frac{\partial(\Delta p)}{\partial t} = R(x) - G(x) \approx \frac{n_i e^{qV/2kT}}{2\tau_0 \cosh\left(\frac{q\mathcal{E}(x_0)}{kT}x\right)}$$

$$\Rightarrow \quad J_{gen} = q\int dx\,[R(x) - G(x)] = \frac{q n_i e^{qV/2kT}}{2\tau_0}\int_{-\infty}^{+\infty} dx\,\frac{1}{\cosh\left(\underbrace{\frac{q\mathcal{E}(x_0)}{kT}}_{a}x\right)}$$

$$\int_{-\infty}^{+\infty}\frac{du}{\cosh(ax)} = \frac{\pi}{a} \quad \Rightarrow \quad J_{gen}(\text{fwd. bias}) \approx \frac{q n_i}{2\tau_0}\left\{\frac{\pi(kT/q)}{\mathcal{E}(x_0)}\right\}\cdot\exp\left(\frac{qV}{2kT}\right) \propto n_i$$

$$\boxed{J_{gen} \propto n_i \exp(qV/2kT)}$$



$$J = \underbrace{J_0}_{\substack{\propto n_i^2 \\ \propto e^{-E_g/kT}}}\left(e^{qV/kT} - 1\right) + \underbrace{J_1}_{\substack{\propto n_i \\ \propto e^{-E_g/2kT}}}\left(e^{qV/2kT} - 1\right)$$

(larger for larger bandgap semiconductors!)

(B) REVERSE BIAS

$$(R - G) \simeq \frac{-n_i^2}{2n_i\tau\cosh\left(\frac{E_t - E_i}{kT}\right)} \simeq -\frac{n_i}{2\tau_0} \Rightarrow J_{gen} = q\int dx\left(-\frac{n_i}{2\tau_0}\right)$$

$$\simeq -\frac{q n_i}{2\tau_0}W \quad \text{(depletion width)}$$

$$\boxed{J_{gen} \simeq \frac{q n_i}{2\tau_0}W}$$

(Thermally generated via traps!)